

Lab 1 - Splunk Dashboard for SSH Logs

Objective

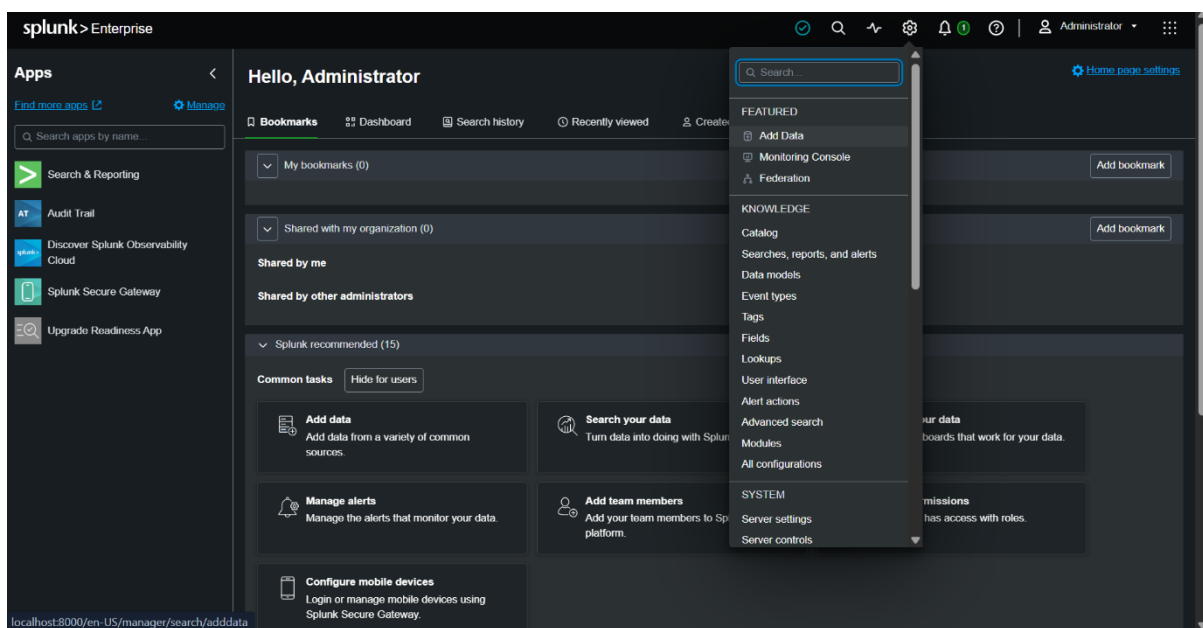
The objective of this project was to monitor SSH authentication activity and identify suspicious login attempts using Splunk dashboards.

For this lab activity, an SSH log dataset was required for analysis. The dataset was obtained from publicly available sources on the internet and saved locally on the system.

Let's Start!

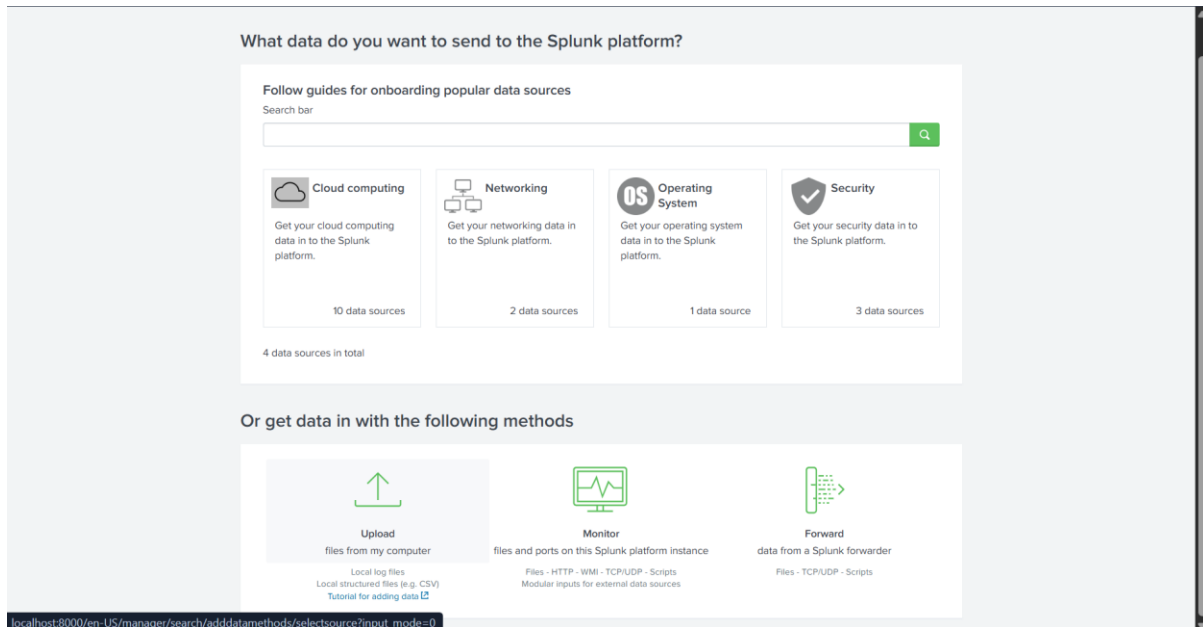
Log in to the Splunk Web interface.

Navigate to Settings → Add Data or click Add Data from the Splunk home page.

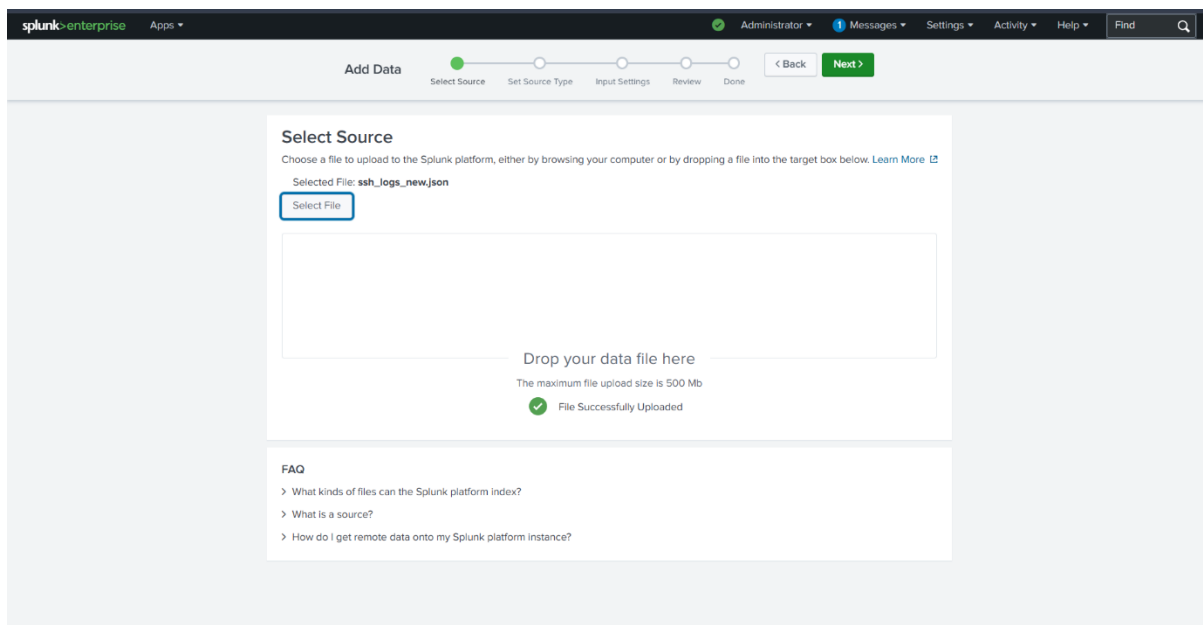


Click on >> Add Data

Select Upload to import data from a local device.



Click On >> Upload



Click On >> Select File

This is our log file

Set Source Type

This page lets you see how the Splunk platform sees your data before indexing. If the events look correct and have the right timestamps, click "Next" to proceed. If not, use the options below to define proper event breaks and timestamps. If you cannot find an appropriate source type for your data, create a new one by clicking "Save As".

Source: `ssh_logs_new.json`

Source type: `_json` Save As

	_time	auth_attempts	auth_success	conn_state	event_type	history	id.orig_h	id.orig_p	id.resp_h	id.re
1	4/24/25 3:50:09.508 PM	1	true	SF	Successful SSH Login	ShADadIF	31.184.137.182	58221	164.254.24.82	22
2	4/24/25 3:50:09.508 PM	1	false	SF	Failed SSH Login	ShADadIF	129.164.50.72	26957	164.186.59.85	22
3	4/24/25 3:50:09.508 PM	8	false	SF	Multiple Failed Authentication Attempts	ShADadIF	135.55.210.223	42848	91.14.145.242	22
4	4/24/25 3:50:09.508 PM	1	false	SF	Failed SSH Login	ShADadIF	125.123.103.182	47789	170.170.25.84	22
5	4/24/25 3:50:09.508 PM	1	true	SF	Successful SSH Login	ShADadIF	4.5.47.4	30192	123.158.160.191	22
6	4/24/25 3:50:09.508 PM	6	false	SF	Multiple Failed Authentication Attempts	ShADadIF	78.164.1.207	32500	91.14.145.242	22
7	4/24/25 3:50:09.508 PM	0	null	SF	Connection Without Authentication	ShADadIF	111.129.222.243	47980	80.101.200.117	22

Click On >> Next

Let's name our host as **"linuxserver"** for easier querying

Input Settings

Optionally set additional input parameters for this data input as follows:

Host

When the Splunk platform indexes data, each event receives a "host" value. The host value should be the name of the machine from which the event originates. The type of input you choose determines the available configuration options. [Learn More](#)

Host field value:

Index

The Splunk platform stores incoming data as events in the selected index. Consider using a "sandbox" index as a destination if you have problems determining a source type for your data. A sandbox index lets you troubleshoot your configuration without impacting production indexes. You can always change this setting later. [Learn More](#)

Index: Default [Create a new index](#)

FAQ

- > How do indexes work?
- > How do I know when to create or use multiple indexes?

Click On >> Review

splunk>enterpriseApps

AdministratorMessagesSettingsActivityHelpFind

Add Data

Select Source

Set Source Type

Input Settings

Review

Done

< BackSubmit >

Review

Input Type Uploaded File
File Name ssh_logs_new.json
Source Type json
Host linuxserver
Index Default

Click on >> Submit

This is our Dashboard

splunk>Enterprise

Administrator

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

6.6

Dashboards

Modules

Dashboards

Visualize and interact with data in one place using Dashboards. Dashboards created outside of Dashboard Studio or Classic have the type Other.

Latest Resources

Examples for Dashboard Studio

Browse examples of dashboards & visualizations. [Visit Example Hub](#)

Intro to Dashboard Studio

Learn how to build dashboards with Dashboard Studio. [Learn More](#)

Intro to Classic Dashboards

Learn how to build traditional Simple XML dashboards. [Learn More](#)

Q filter

This app's

Show only favorites

6 Dashboards

Show: 50 per page

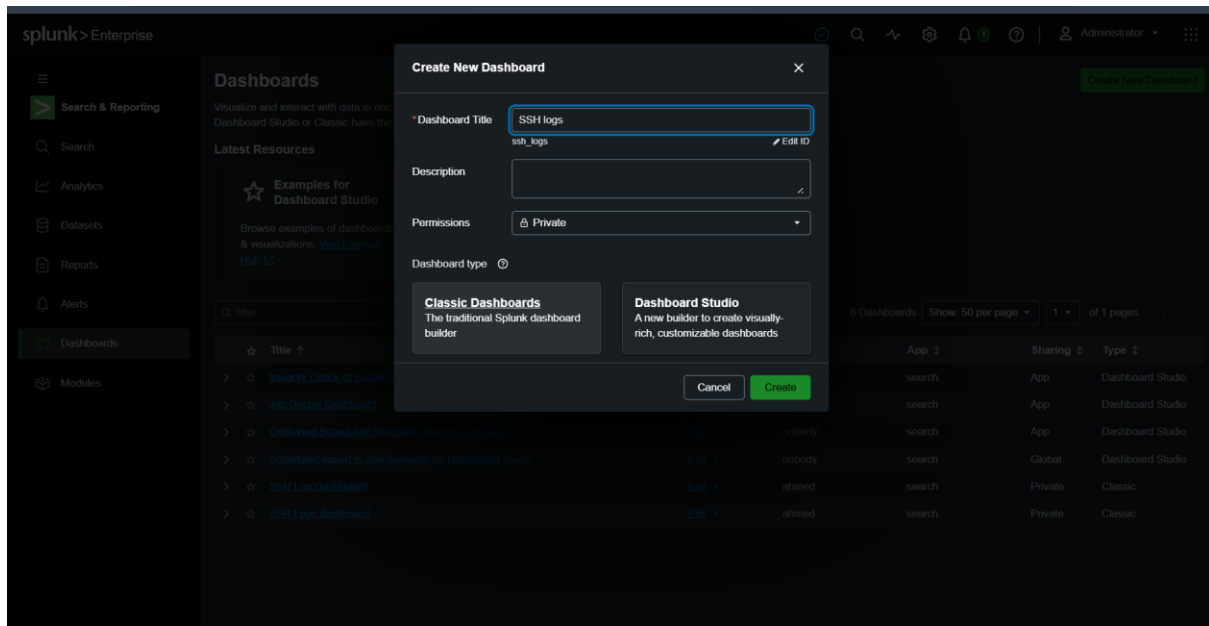
1

of 1 pages

☆	Title ↑	Actions	Owner ↕	App ↕	Sharing ↕	Type ↕
> ☆	Integrity Check of Installed Files	Edit	nobody	search	App	Dashboard Studio
> ☆	Job Details Dashboard	Edit	nobody	search	App	Dashboard Studio
> ☆	Orphaned Scheduled Searches, Reports, and Alerts	Edit	nobody	search	App	Dashboard Studio
> ☆	Scheduled export is now available for Dashboard Studio	Edit	nobody	search	Global	Dashboard Studio
> ☆	SSH Log Dashboard	Edit	ahmed	search	Private	Classic
> ☆	SSH Logs dashboard	Edit	ahmed	search	Private	Classic

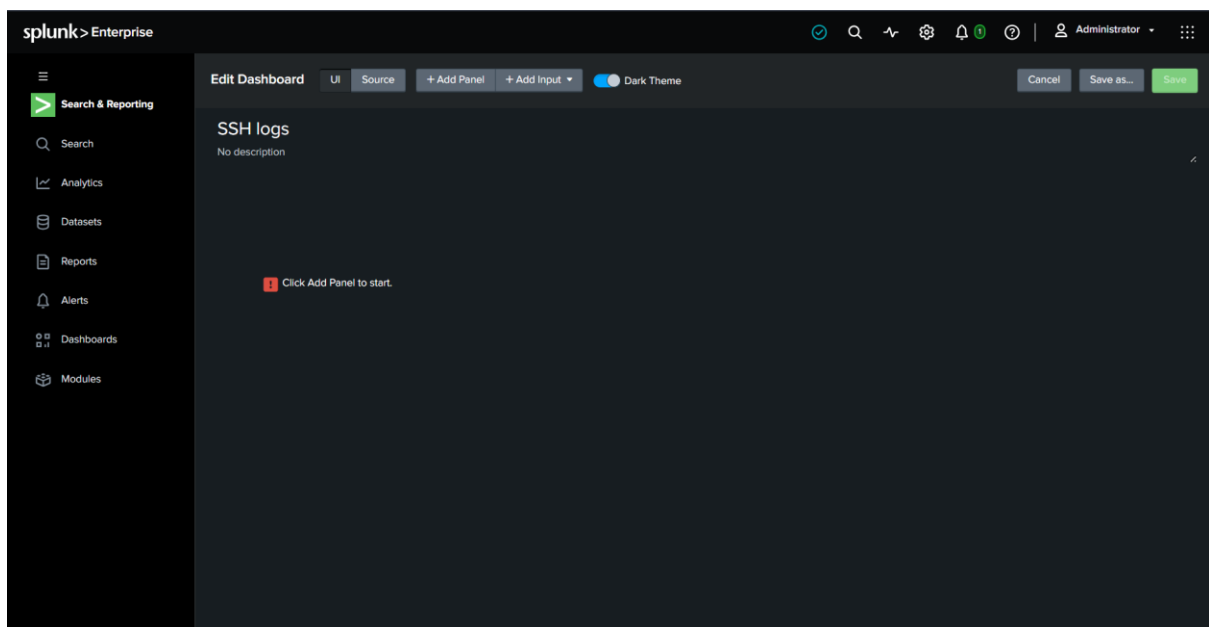
Click On >> Create New Dashboard

Let us name our dashboard as SSH logs, we will select Classic Dashboard as Dashboard Type

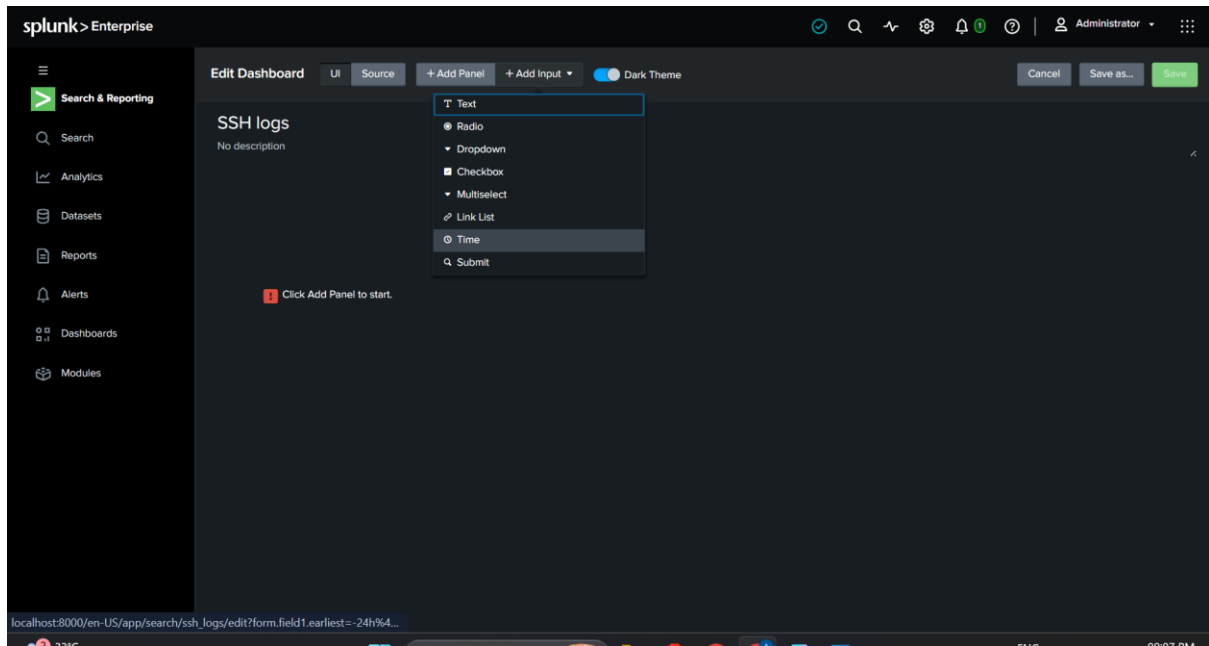


Click on >> Create

This is how our dashboard looks now, at first we need to setup time range and add submit button

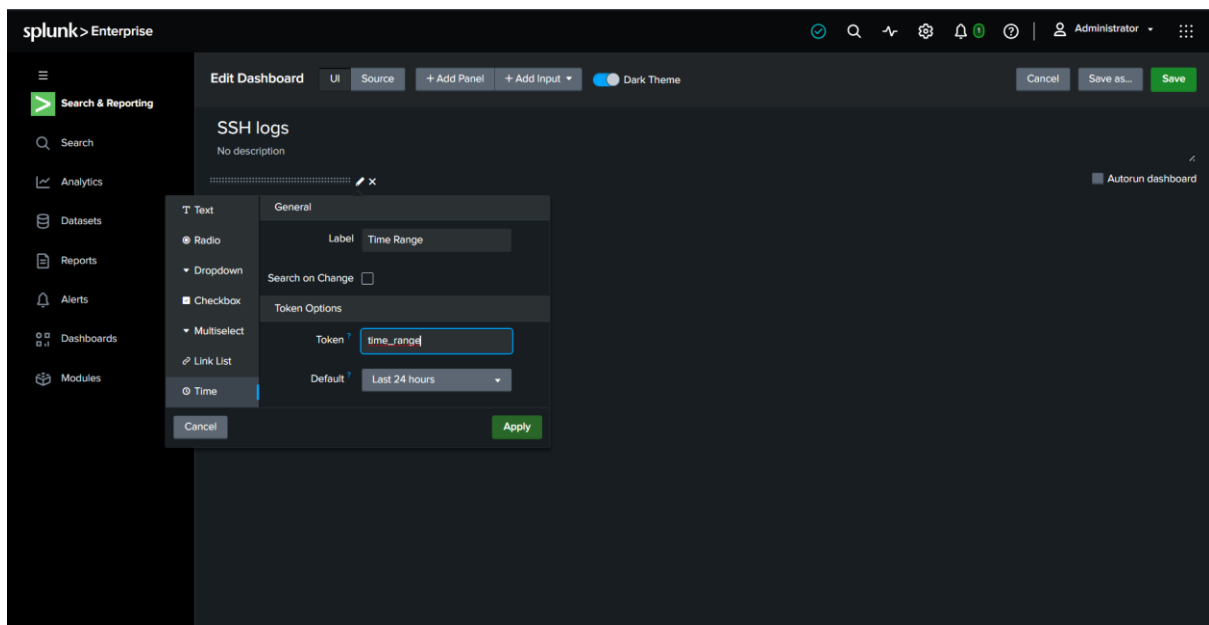


Let us setup time range first



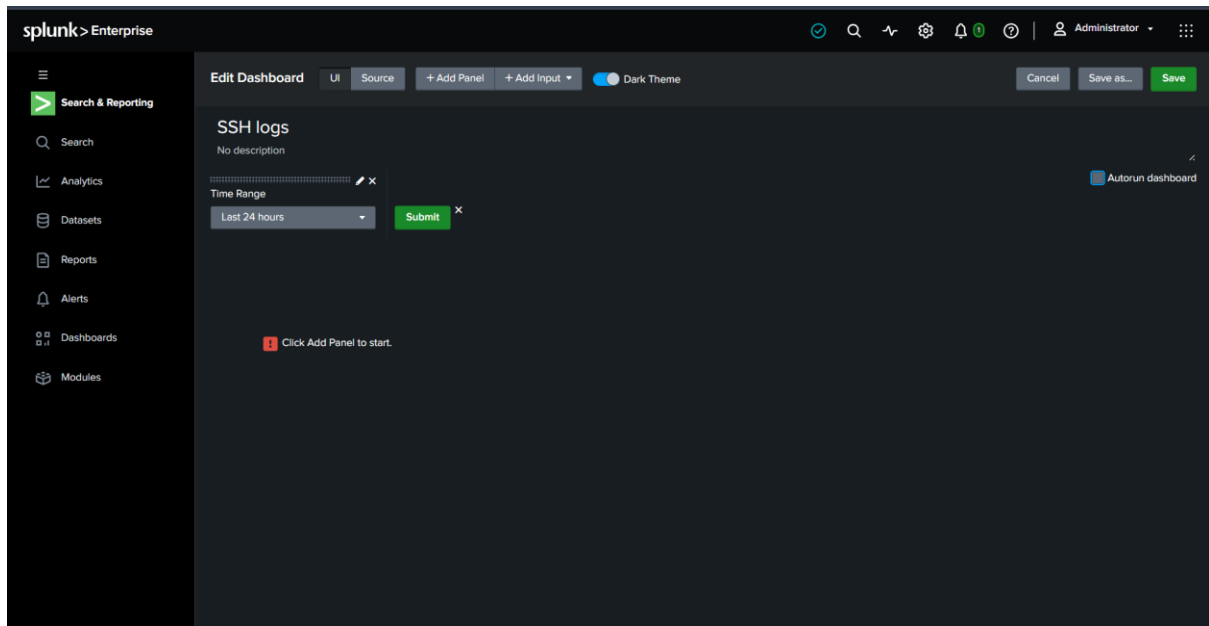
Click on >> Add Input >> Time

Click on pencil icon, Set Label to Time Range and Token time_range



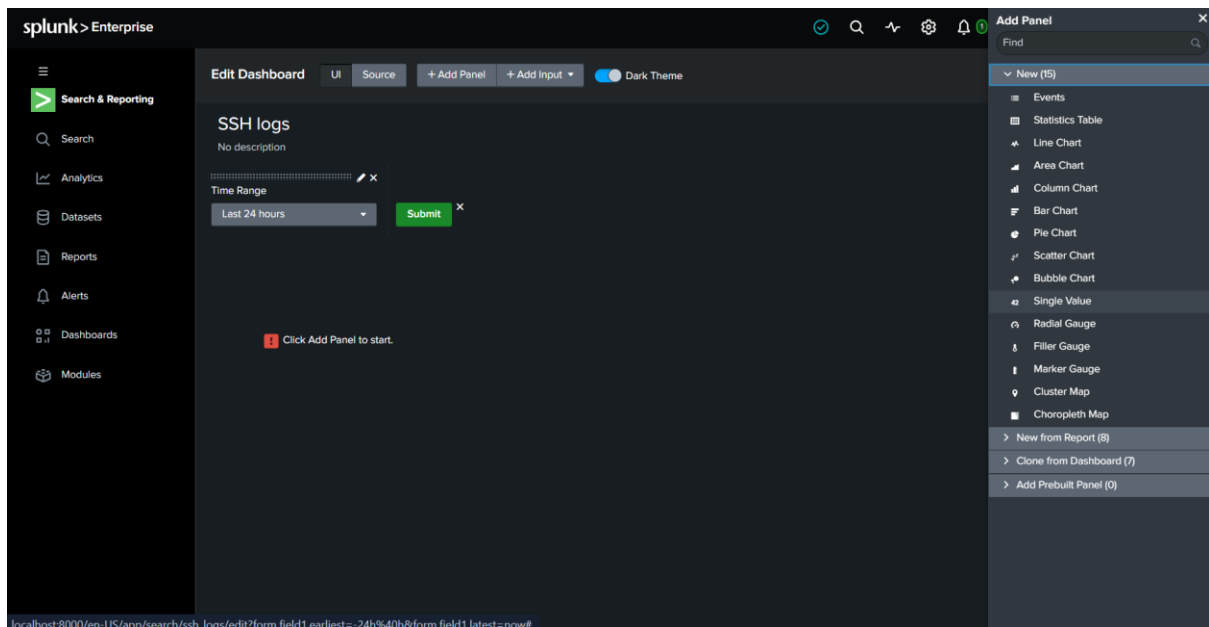
Click On >>Select Submit

Similarly add Submit button



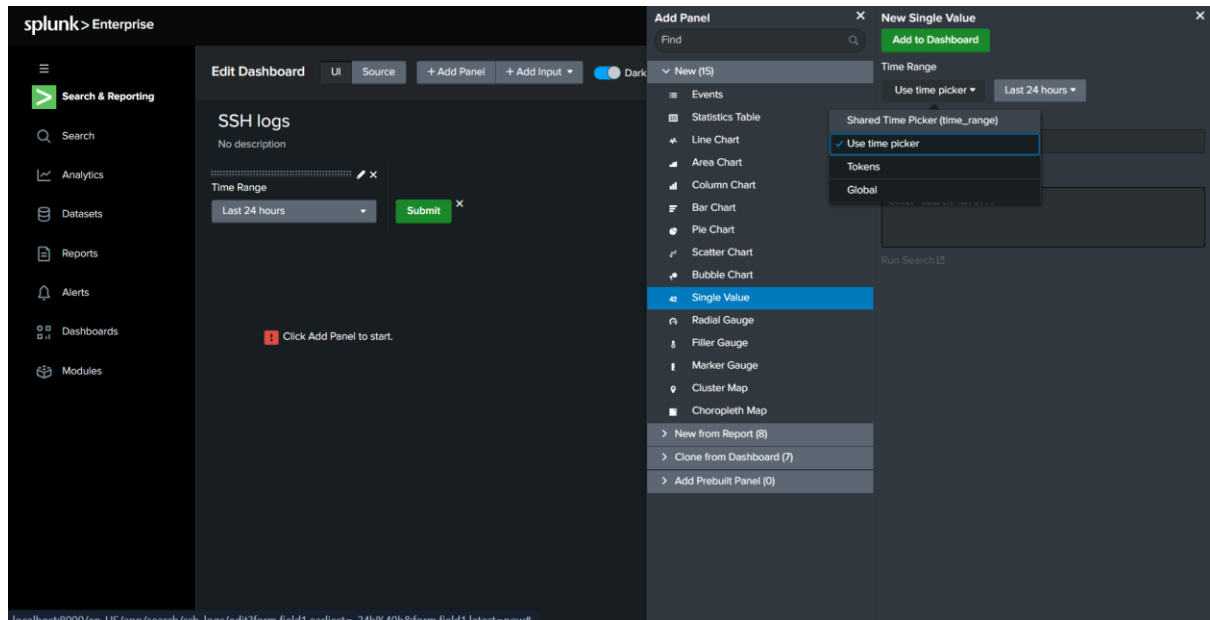
1. Total SSH Events

- Click on Add Panel
- Under New, choose Single Value



Use Shared Time Picker **time_range**

Set Content Title to "Total SSH Events"



Enter the Search String as below

```
source="ssh_logs_new.json" host="linuxServer"
sourcetype="_json" | stats count AS "Total SSH Events"
```

Explanation

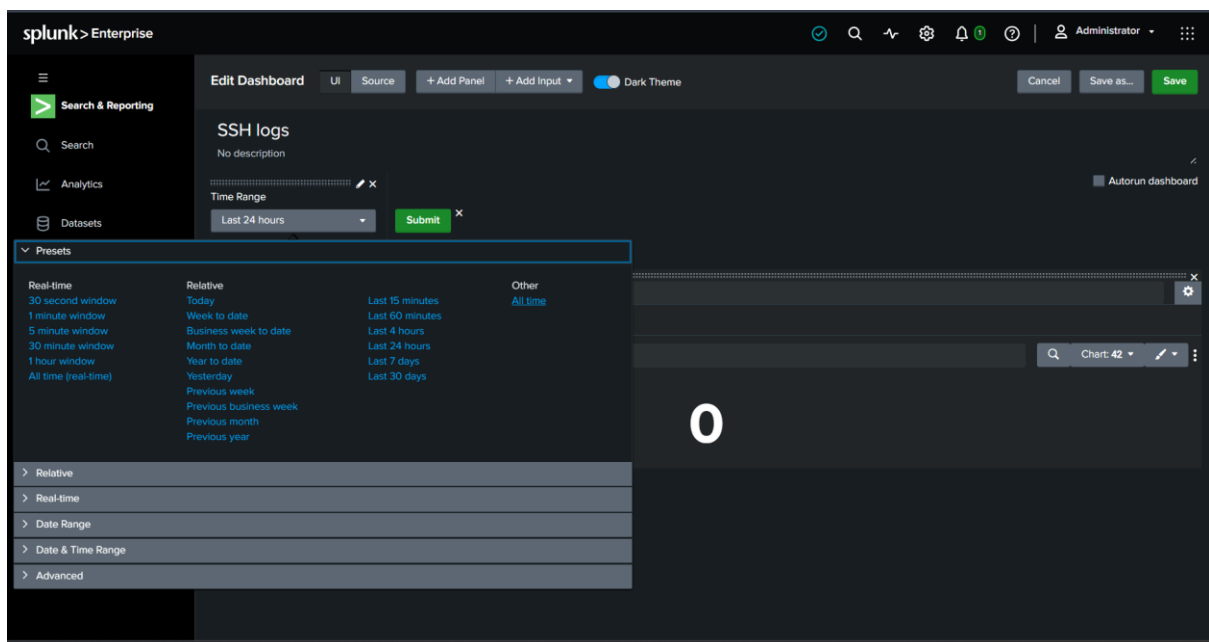
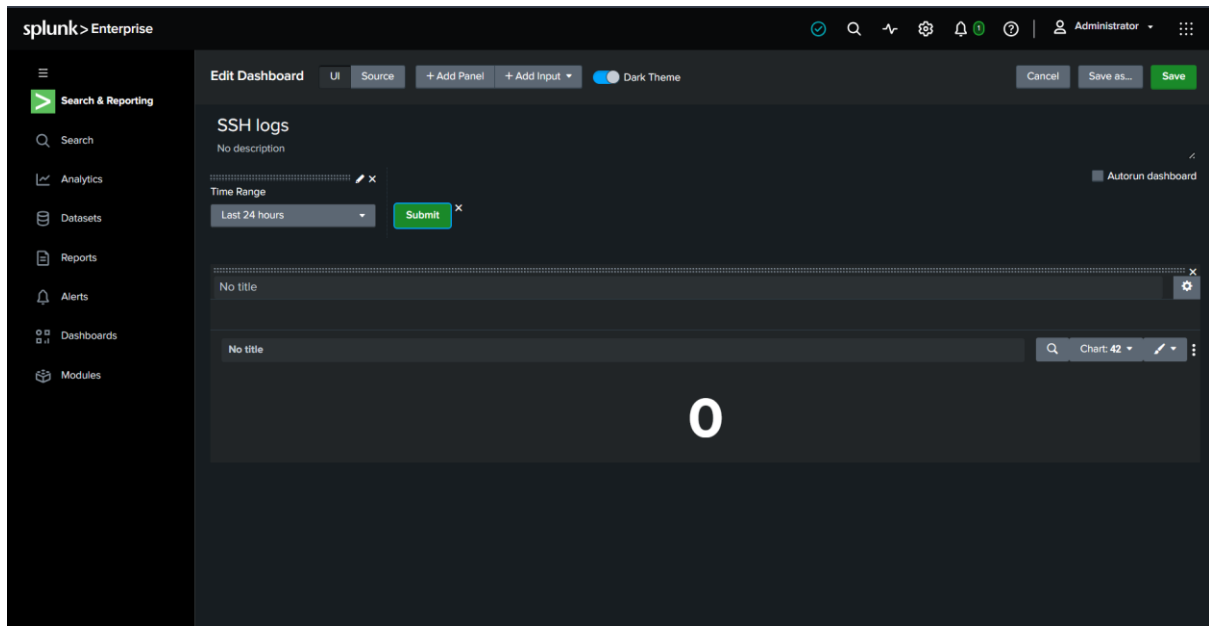
This query searches the SSH log dataset stored in Splunk and filters events based on the specified source file, host, and source type.

- `source="ssh_logs_new.json"` filters events originating from the SSH log file named **ssh_logs_new.json**.
- `host="linuxServer"` restricts the search to logs generated by the host **linuxServer**.
- `sourcetype="_json"` tells Splunk to search only JSON-formatted events.

The stats command is then used to perform statistical analysis on the filtered data:

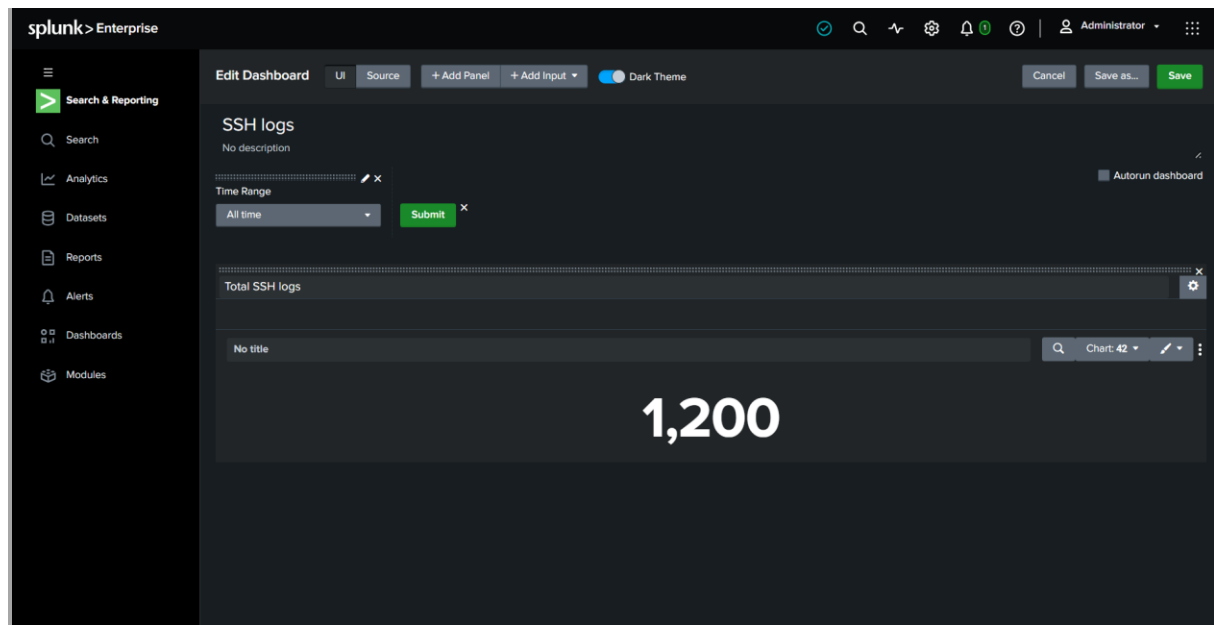
- `count` calculates the total number of events matching the search criteria.
- `AS "Total SSH Events"` renames the output field to **Total SSH Events** for better readability in reports and dashboards.

We don't see the result, its because the logs are from past so, let us change time to **All Time**



Click On >> Submit

This is how the dashboard look like, change the Time Range to “All Time”



Similarly let us find out Successful Logins, Failed Logins

2. Successful Logins

- Click on **Add Panel**
- Under **New**, choose **Single Value**
- Use **Shared Time Picker time_range**
- Set **Content Title** to "Successful Logins"
- Enter the **Search String** as below:

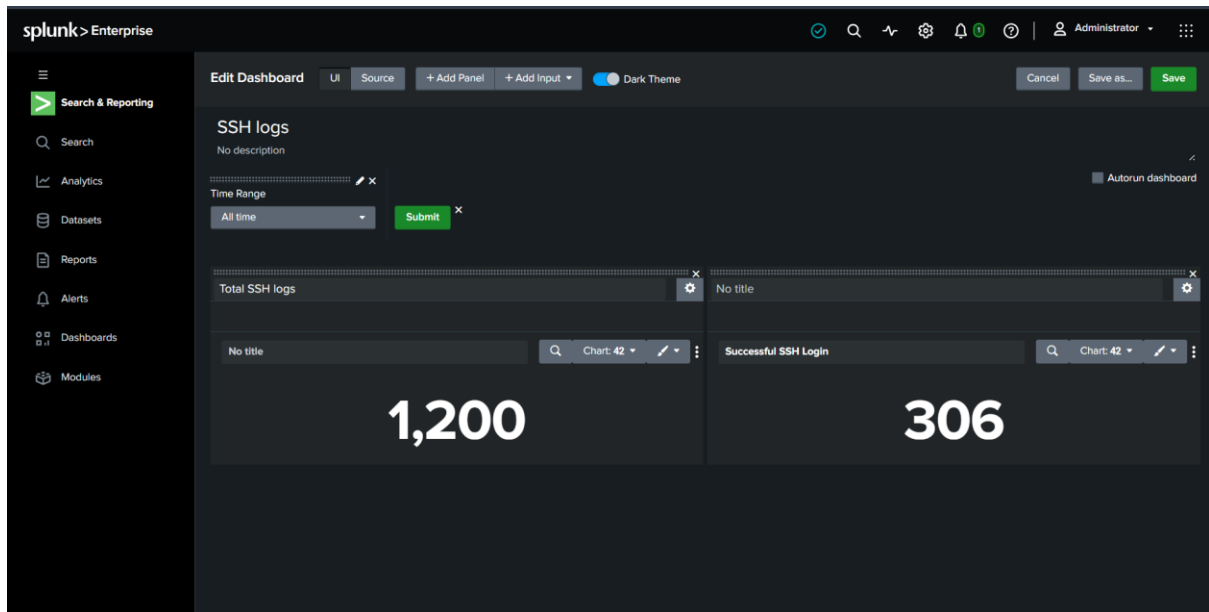
```
source="ssh_logs_new.json" host="linuxServer" sourcetype="_json"
event_type="Successful SSH Login"
| stats count AS "Successful Logins"
```

Explanation

This query searches the SSH log dataset and filters events related to successful SSH authentication attempts.

- `source="ssh_logs_new.json"` searches events originating from the specified SSH log file.
- `host="linuxServer"` limits the search to logs generated by the Linux server.
- `sourcetype="_json"` filters for events stored in JSON format.

- `event_type="Successful SSH Login"` retrieves only events where a user successfully authenticated via SSH.



3. Failed Logins

- Click on **Add Panel**
- Under **New**, choose **Single Value**
- Use **Shared Time Picker** `time_range`
- Set **Content Title** to **"Failed Logins"**
- Enter the **Search String** as below:

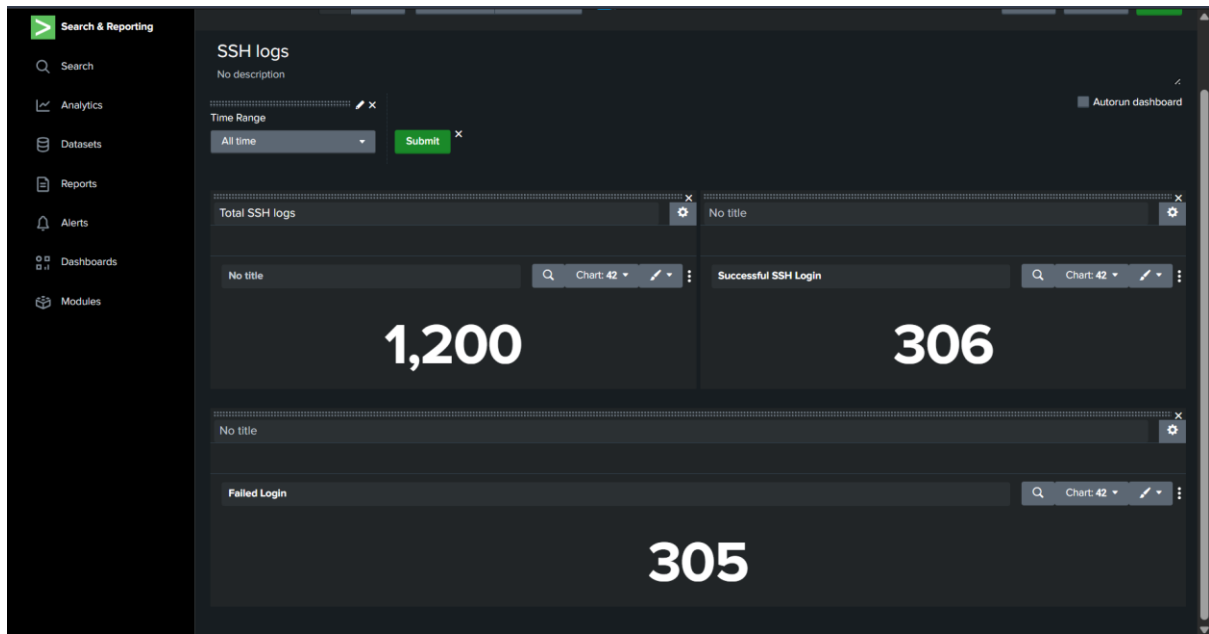
```
source="ssh_logs_new.json" host="linuxServer"
sourcetype="_json" event_type="Failed SSH Login"
| stats count AS "Failed Login"
```

Explanation

This query searches the SSH log dataset and identifies all failed SSH authentication attempts. It then counts the total number of failed login events.

- `source="ssh_logs_new.json"` filters events from the specified SSH log file.
- `host="linuxServer"` limits the search to logs generated by the Linux server.
- `sourcetype="_json"` searches only JSON-formatted log events.

- event_type="Failed SSH Login" filters for failed SSH authentication attempts.
- stats count calculates the total number of matching events.
- AS "Failed Login" renames the output field to **Failed Login** for easier interpretation in dashboards and reports.



4. Connection without Authentication

- Click on **Add Panel**
- Under **New**, choose **Single Value**
- Use **Shared Time Picker time_range**
- Set **Content Title** to "Invalid User Attempts"
- Enter the **Search String** as below:

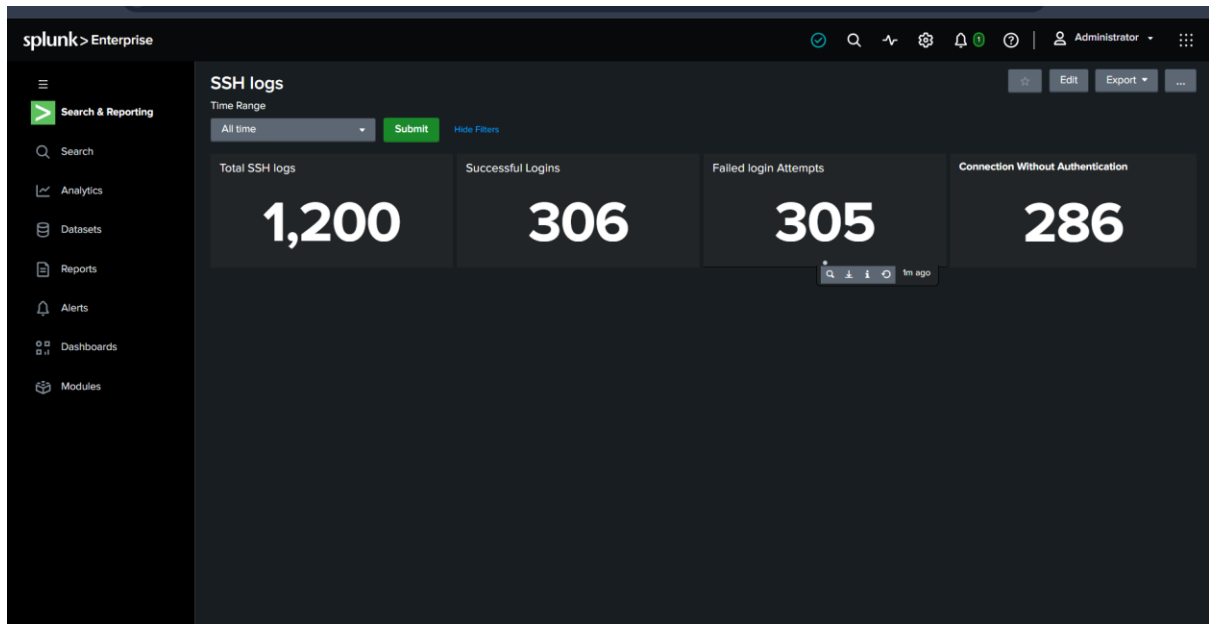
```
source="ssh_logs_new.json" host="linuxServer"
sourcetype="_json" event_type="Connection Without
Authentication"
| stats count AS "Connection Without Authentication"
```

Explanation

This query identifies SSH connections that were established but did not proceed to successful authentication.

- source="ssh_logs_new.json" searches events from the SSH log dataset.

- `host="linuxServer"` filters events generated by the Linux server.
- `sourcetype="_json"` restricts the search to JSON-formatted logs.
- `event_type="Connection Without Authentication"` retrieves events where a connection attempt occurred without successful user authentication.
- `stats count` calculates the total number of matching events.



5. Failed Login by Username

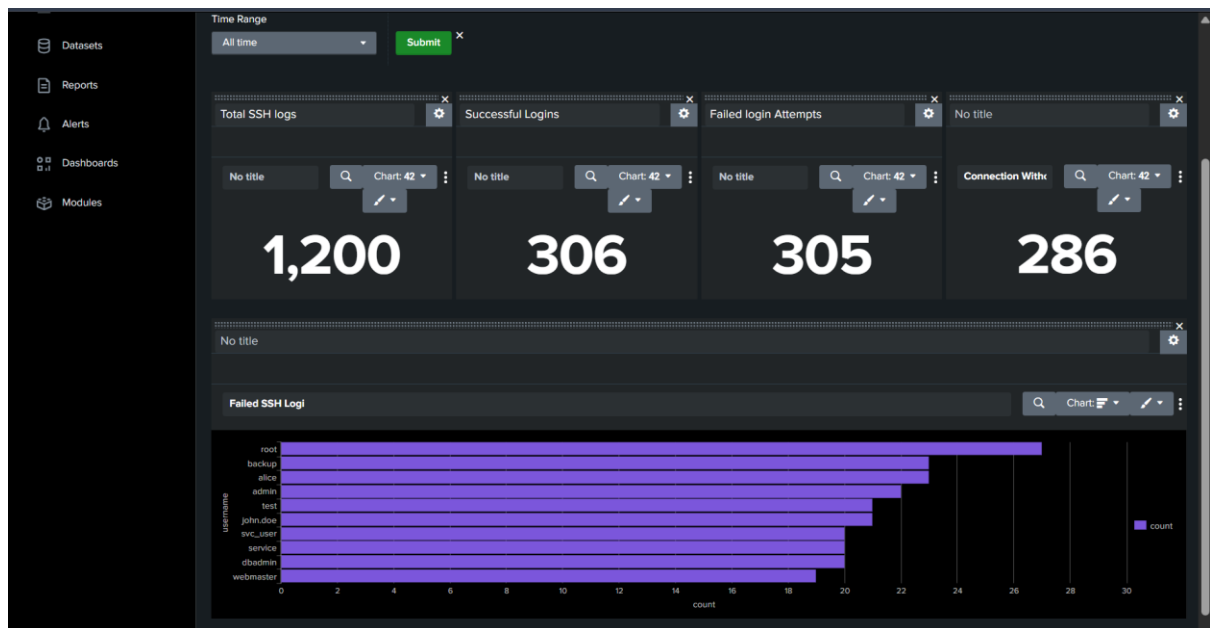
- Click on **Add Panel**
- Under **New**, choose **Bar Chart**
- Use **Shared Time Picker** `time_range`
- Set **Content Title** to **"Failed Logins by username"**
- Enter the **Search String** as below:

```
source="ssh_logs_new.json" host="linuxServer"
sourcetype="_json" event_type="Failed SSH Login"
| top username
```

Explanation

This query identifies the usernames that experience the highest number of failed SSH login attempts.

- `event_type="Failed SSH Login"` filters failed authentication events.
- `top username` displays the usernames with the highest frequency of failed login attempts.



6. Possible Brute Force

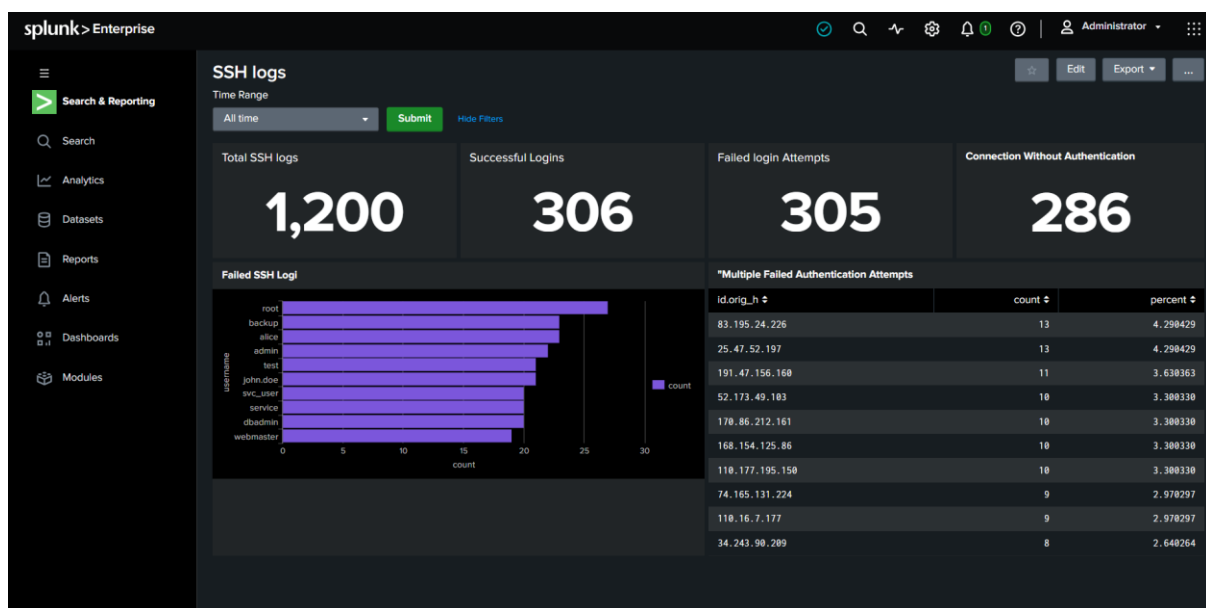
- Click on **Add Panel**
- Under **New**, choose **Statistics Table**
- Use **Shared Time Picker** `time_range`
- Set **Content Title** to **Possible Brute Force b IP Address**
- Enter the **Search String** as below:

```
source="ssh_logs_new.json" host="linuxServer"  
sourcetype="_json" event_type="Multiple Failed Authentication Attempts" | top id.orig_h
```

Explanation

This query identifies source IP addresses responsible for multiple failed authentication attempts.

- `event_type="Multiple Failed Authentication Attempts"` filters events associated with repeated login failures.
- `top id.orig_h` displays the IP addresses generating the highest number of failed login attempts.



7. Visualizing Brute Force attack with geo-location

- Click on **Add Panel**
- Under **New**, choose **Choropleth Map**
- Use **Shared Time Picker** `time_range`
- Set **Content Title** to **Brute Force attack with geo-location**

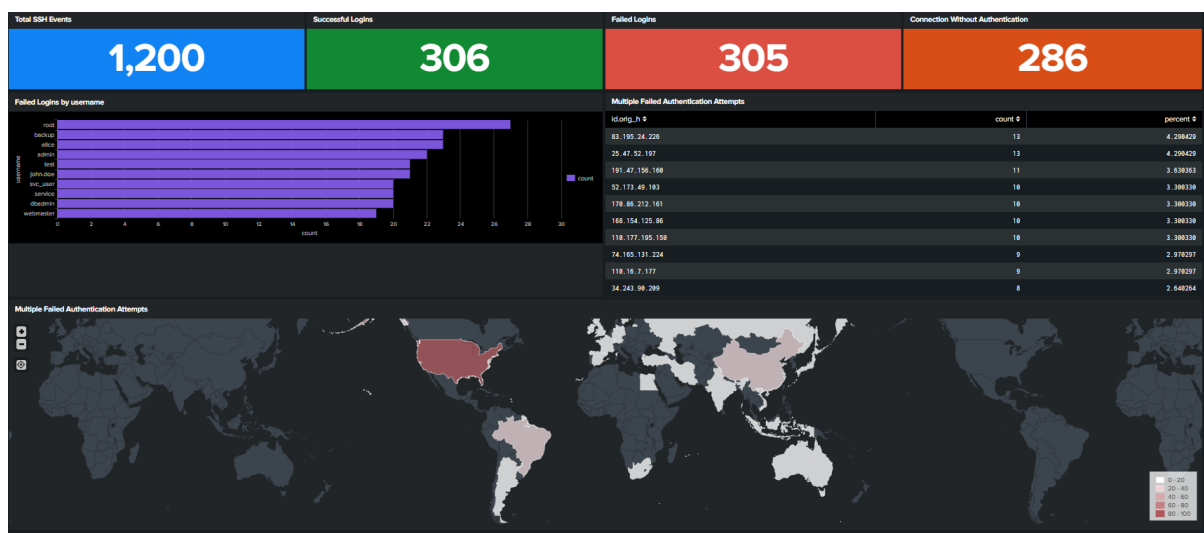
Enter the **Search String** as below:

```
source="ssh_logs_new.json" host="linuxServer"
sourcetype="_json" event_type="Multiple Failed Authentication Attempts"
| table id.orig_h
| iplocation id.orig_h
| stats count by Country
| geom geo_countries featureIdField="Country"
```

Explanation

This query visualizes the geographical origins of IP addresses involved in multiple failed SSH authentication attempts.

- table id.orig_h extracts the source IP addresses from the events.
- iplocation id.orig_h performs geolocation lookup and retrieves location information such as country, city, and region.
- stats count by Country counts the number of suspicious events originating from each country.
- geom geo_countries featureIdField="Country" maps the results onto a world map for visualization.



Conclusion

This lab successfully demonstrated how to build an SSH log monitoring dashboard in Splunk Enterprise. By ingesting a JSON-formatted SSH log dataset, the dashboard provided four key metrics Total SSH Events (1,200), Successful Logins (306), Failed Logins (305), and Connections Without Authentication (286). The high failure rate and unauthenticated connections indicate active brute-force activity, further confirmed by the geo-location map showing the geographic origins of attacks. Overall, this lab highlighted the practical value of Splunk as a SIEM tool for detecting and visualizing SSH-based threats in a SOC.